ITAI-1371-Intro to Machine Learning-RT-14274

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Team ML_1371_14274 2

Lab 08 Reflective Journal

Bias-Variance Tradeoff

This lab helped our team understand why creating a more complex machine learning model does not automatically mean creating a better model. Before completing this exercise, we understood that a model should learn patterns from data, but the visual examples made it much clearer how a model can either learn too little or learn too much. Comparing different polynomial models on the same dataset helped us better understand the concepts of underfitting, overfitting, bias, and variance.

The lab used a synthetic dataset based on a sine wave with added noise. This was useful because the general pattern of the data was visible, while the individual observations did not perfectly follow the curve. This helped us connect the exercise to real-world machine learning problems, where data often contains noise and irregularities. A good model should learn the important pattern in the data without trying to reproduce every individual observation.

One of the most important parts of the lab was comparing Polynomial Regression models with degrees 1, 4, and 15. The degree 1 model was too simple because it could only represent a basic linear relationship. It could not capture the curved pattern of the data. This demonstrated underfitting and high bias. The model makes assumptions that are too simple for the actual relationship and therefore performs poorly even on the training data.

The degree 4 model demonstrated a better balance between simplicity and complexity. It was flexible enough to follow the general shape of the data without reacting too strongly to the random noise. This helped us understand that the goal of machine learning is not necessarily to create a model that perfectly matches every training observation. Instead, the goal is to learn a pattern that can also perform well when presented with new data.

The degree 15 model demonstrated the opposite problem. The model became highly flexible and began following small variations and noise in the training data. This helped us visualize overfitting and high variance. A model can appear to perform extremely well on its training

data while still performing poorly when it receives new information. This showed us why training performance by itself is not enough to determine whether a model is actually good.

The learning curves provided another useful way to understand these problems. For the simpler model, the training and cross-validation results showed that the model could not learn the underlying pattern well. For the more complex model, the difference between training and validation performance showed that the model was learning the training data much better than it was generalizing to unseen data. Seeing these results made the concepts of high bias and high variance easier to recognize.

This lab also reinforced concepts from our previous exercises involving training and testing data. We now better understand why machine learning models must be evaluated using information that was not used during training. A model may memorize patterns in the training set, but this does not guarantee that those patterns will be useful when predicting new observations. Validation and testing allow us to measure the model's ability to generalize rather than simply memorize.

One of the main concepts our team learned from this lab was the meaning of the bias-variance tradeoff. A model with too much bias is too simple and may underfit the data, while a model with too much variance becomes too sensitive to the training data and may overfit. The best solution is usually somewhere between these two extremes, where the model is complex enough to learn the important patterns but still simple enough to perform well on new data.

We can also connect this concept to our Real Estate machine learning project. When we begin training models to predict Sale Amount, a model that performs extremely well on the training dataset will not automatically be the best model. We will also need to compare its performance on unseen data and determine whether it is learning meaningful relationships or simply memorizing characteristics of the training dataset.

Overall, this lab changed the way our team thinks about model complexity. We now understand that increasing model complexity can improve performance only up to a certain point. After that point, additional complexity can reduce the model's ability to generalize. The comparison between degree 1, degree 4, and degree 15 models made the bias-variance tradeoff much more concrete and helped us understand why finding the right balance is an important part of building effective machine learning models.